



Assignment 2

TFB2063: Data Science

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1.0 Introduction

1.1 Problem Scenario

In the competitive landscape of modern banking, customer acquisition strategies must evolve from broad, untargeted campaigns to data-driven, personalized marketing. Our client, a rapidly growing young bank (referred to as JB Bank for this analysis), faces a common challenge: a robust base of liability customers (depositors) but a comparatively small portfolio of asset customers (borrowers). To drive revenue growth, the bank is keen on expanding its loan business by converting its existing depositors into personal loan customers. A previous marketing campaign yielded a healthy conversion rate of over 9%, demonstrating a significant opportunity. This success has motivated the retail marketing department to refine its approach. Rather than marketing to all liability customers, the bank aims to use predictive analytics to identify the specific individuals most likely to accept a personal loan offer. This targeted method promises to increase campaign efficiency, reduce marketing costs and improve customer experience by offering relevant products. The core challenge, therefore, is to analyze the historical customer data from the previous campaign to build a model that can accurately predict which liability customers will become personal loan holders. The insights derived from this analysis will form the foundation for designing a smarter, more effective future campaign.

1.2 Project Objective

This project has two primary analytical objectives, designed to provide actionable insights for the bank's retail marketing strategy:

1. **Predictive Modeling for Loan Acceptance (Classification):** The primary goal is to develop a robust predictive model to classify customers based on their likelihood of accepting a personal loan. By analyzing the profiles and behaviors of customers from the previous campaign, we aim to identify the key factors that influence a customer's decision. This will enable the bank to prioritize outreach to high-potential prospects.
2. **Identifying Cross-Selling Opportunities (Association Analysis):** Beyond predicting loan acceptance, the bank is interested in understanding the relationships between its various financial products. This objective focuses on uncovering associations among product holdings such as Certificate of Deposit (CD) accounts, securities accounts and credit cards among customers who are predicted to accept a loan. Identifying these patterns will reveal valuable cross-selling opportunities, allowing the bank to offer complementary products and deepen its relationship with new loan customers.

1.3 Dataset Overview

The dataset provided for this analysis contains historical information for 5,000 bank customers. This data includes demographic profiles, details of their relationship with the bank and their response to the last personal loan campaign. The target variable for our predictive model is clearly identified.

Key attributes in the dataset include:

- **Demographics:** Age, annual Income (in \$000s), Family size, and Education level.
- **Customer Relationship:** Years of professional Experience, average monthly spending on credit cards (CCAvg), and value of any existing Mortgage (in \$000s).
- **Product Holdings:** Indicators for various bank products, including Personal Loan (the target variable for this study), Securities Account, Certificate of Deposit (CD) Account, Online banking usage, and Credit Card issued by the bank.
- **Non-Predictive Identifiers:** ID and ZIP Code, which are present for record-keeping but are logically irrelevant for predicting customer behavior and will be removed during the data preparation phase.

This rich dataset provides a comprehensive view of the customer base, offering a solid foundation for building an accurate and insightful predictive model.

2.0 Data Cleaning and Data Partitioning

2.1 Negative Experience Value Issue

During the data exploration phase, it was identified that the Experience variable contained several negative values. As this variable reflects the number of years of professional experience for a customer, negative values are logically invalid and lack real-world significance. This inconsistency is likely attributable to data entry or recording errors.

If these negative values are left unaddressed, these values could adversely impact the performance of predictive models, especially for distance-based algorithms such as k-NN, where inaccurate values can distort similarity calculations between customers.

2.2 Data Cleaning Process

To address this issue, the negative values in the Experience variable were transformed into their absolute values. This remedial measure is predicated on the assumption that the sign error occurred during data entry, while the numeric magnitude of the values remains valid.

In addition to addressing these invalid entries, basic data cleaning procedures were implemented to ensure the dataset is suitable for analysis. This includes ensuring that all variables are clean, accurate, and consistent across all data. Irrelevant attributes, such as ID and ZIP Code, previously identified in Section 1.3, are retained at this juncture but will be eliminated during the feature selection stage to prevent any undue influence on the prediction model.

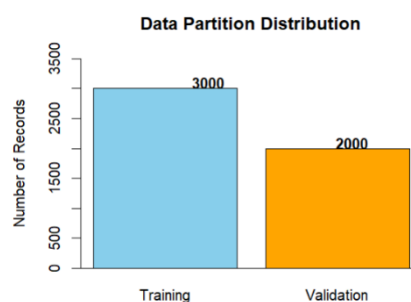
By executing these data cleaning steps, the dataset attains a higher degree of accuracy and reliability, which is essential for building an effective predictive model.

2.3 Data Partitioning Method (60% Training / 40% Validation)

After completing the data cleaning process, the dataset was segmented into two distinct subsets: a training set and a validation set, reflecting a split of 60% and 40%, respectively. This division is critical for facilitating an objective assessment of the model's efficacy. The training set, comprising 60% of the dataset, serves to instruct the k-Nearest Neighbors (k-NN) algorithm on the relationships between predictor variables and the target variable, Personal Loan. During this training phase, the model identifies patterns that differentiate customers who accepted the loan from those who declined.

The remaining 40% of the dataset is designated as the validation set, which is employed to evaluate the model's performance on previously unseen data. This step is vital for determining how well the model generalizes beyond the training dataset. If the model performs well on training data but poorly on validation data, it may indicate overfitting. Thus, the use of a separate validation set provides a more accurate reflection of predictive performance.

The 60:40 split strikes a commendable balance between training and evaluation. It guarantees an adequate amount of data for effective learning while offering a reliable foundation for testing the model's accuracy. Furthermore, this methodology enables further analyses, such as exploring various values of k within the k-NN algorithm, thereby enhancing the development of a robust predictive model.



3.0 Feature Selection and k-NN Implementation

3.1 Feature Selection Strategy

The first step in building a strong prediction model is to ensure that only relevant information is fed into the algorithm. Including irrelevant attributes or 'noise' can confuse the model and reduce its performance. We eliminated two attributes from the dataset before modelling:

1. ID: This is a unique sequential identifier for each customer row. It has no intrinsic relationship with a customer's likelihood of accepting loan. In distance-based algorithms like k-NN, including such an arbitrary number but unique will force the model to treat all customers as different.
2. ZIP Code: ZIP code is the categorical identifier. Its values are arbitrary and do not represent a meaningful similarity for likelihood of customer accepting loan. Including it will be causing noise and distort distance calculations.

The following R code was used to remove these attributes.

```
> HypotheticalBank$ID <- NULL
```

```
> HypotheticalBank$ ZIP Code <- NULL
```

Before deleting:

```
> str(HypotheticalBank)
tibble [5,000 × 14] (S3: tbl_df/tbl/data.frame)
 $ ID          : num [1:5000] 1 2 3 4 5 6 7 8 9 10 ...
 $ Age         : num [1:5000] 25 45 39 35 35 37 53 50 35 34 ...
 $ Experience   : num [1:5000] 1 19 15 9 8 13 27 24 10 9 ...
 $ Income      : num [1:5000] 49 34 11 100 45 29 72 22 81 180 ...
 $ ZIP Code    : num [1:5000] 91107 90089 94720 94112 91330 ...
 $ Family      : num [1:5000] 4 3 1 1 4 4 2 1 3 1 ...
 $ CCAvg       : num [1:5000] 1.6 1.5 1 2.7 1 0.4 1.5 0.3 0.6 8.9 ...
 $ Education   : num [1:5000] 1 1 2 2 2 2 3 2 3 ...
 $ Mortgage    : num [1:5000] 0 0 0 0 155 0 0 104 0 ...
 $ Personal Loan : num [1:5000] 0 0 0 0 0 0 0 0 1 ...
 $ Securities Account: num [1:5000] 1 1 0 0 0 0 0 0 ...
 $ CD Account  : num [1:5000] 0 0 0 0 0 0 0 0 ...
 $ Online      : num [1:5000] 0 0 0 0 1 1 0 1 0 ...
 $ CreditCard  : num [1:5000] 0 0 0 0 1 0 0 1 0 ...
```

After deleting:

```
> str(HypotheticalBank)
tibble [5,000 × 12] (S3: tbl_df/tbl/data.frame)
 $ Age         : num [1:5000] 25 45 39 35 35 37 53 50 35 34 ...
 $ Experience   : num [1:5000] 1 19 15 9 8 13 27 24 10 9 ...
 $ Income      : num [1:5000] 49 34 11 100 45 29 72 22 81 180 ...
 $ Family      : num [1:5000] 4 3 1 1 4 4 2 1 3 1 ...
 $ CCAvg       : num [1:5000] 1.6 1.5 1 2.7 1 0.4 1.5 0.3 0.6 8.9 ...
 $ Education   : num [1:5000] 1 1 2 2 2 2 3 2 3 ...
 $ Mortgage    : num [1:5000] 0 0 0 0 155 0 0 104 0 ...
 $ Personal Loan : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 2 ...
 $ Securities Account: num [1:5000] 1 1 0 0 0 0 0 0 ...
 $ CD Account  : num [1:5000] 0 0 0 0 0 0 0 0 ...
 $ Online      : num [1:5000] 0 0 0 0 1 1 0 1 0 ...
 $ CreditCard  : num [1:5000] 0 0 0 0 1 0 0 1 0 ...
```

3.2 K-Nearest Neighbour (k-NN) Algorithm

k-NN is a simple classification algorithm. It predicts the class for new customer by looking at the 'k' most similar customer in the training data.

How it works:

1. Calculate Distance: It measures the distance like Euclidean between the new customer and every customer in the training set.
2. Find Neighbours: It selects the 'k' training customers with the smallest distances.
3. Vote for Class: It counts how many of these neighbours accepted the personal loan (Class 1) versus those who did not (Class 0).
4. Predict: The new customer is assigned the class that has the majority vote.

3.3 Data Normalization

k-NN performance depends on its distance calculation. If the features have different scales (Income range 8-244 vs. Education range 1-3), Income will dominate the distance, causing the model to be biased. The solution of Normalization will scale all numeric features to a standard range (0 to 1) ensuring each contributes equally.

Normalization Formula and Code to do Formalization:

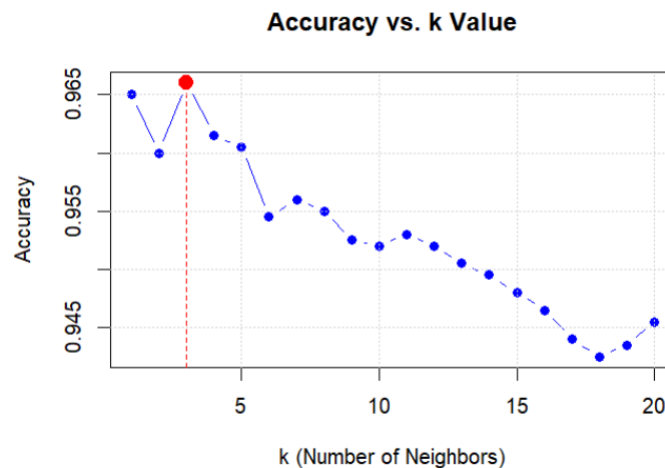
```
> normalize <- function(x){
+   +   return((x - min(x)) / (max(x) - min(x))) }
+
> train_norm <- as.data.frame(apply(train_data[, -which(names(train_data)=="Personal Loan")], normalize))
> valid_norm <- as.data.frame(apply(valid_data[, -which(names(valid_data)=="Personal Loan")], normalize))
>
> train_target <- train_data$Personal Loan
>
> valid_target <- valid_data$Personal Loan
> install.packages("class")
```

By selecting only the relevant features then normalize them, we have prepared a clean and fair dataset for the k-NN algorithm. The next step is to find the optimal value of 'k'.

4.0 Optimal k Selection and Model Evaluation

4.1 Testing Different k Values

Different values of k ranging from **1 to 20** were tested using the validation dataset. For each value of k , the k-NN model was applied and the corresponding classification accuracy was calculated. The results show that the **highest accuracy was obtained when $k = 3$** . If k is too small (e.g., $k = 1$), the model becomes sensitive to noise and outliers. If k is too large, the neighbourhood may include too many observations from different classes, which reduces prediction accuracy. Therefore, **$k = 3$ was selected as the optimal value** because it provides a good balance between model sensitivity and generalization.

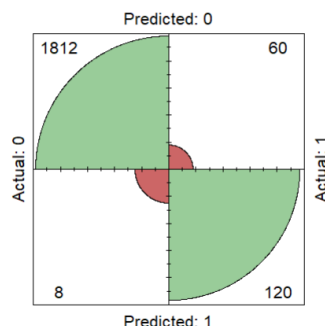


4.2 Confusion Matrix

After selecting the optimal value $k = 3$, the model performance was evaluated using a **confusion matrix** on the validation dataset. The confusion matrix compares the **actual values** with the **predicted values** produced by the k-NN model.

It consists of four outcomes: **True Positive (TP)**, **True Negative (TN)**, **False Positive (FP)**, and **False Negative (FN)**, which help evaluate how well the model classifies customers who accept or reject the loan.

Confusion Matrix: Prediction Accuracy



4.3 Model Accuracy

Model accuracy represents the proportion of correct predictions made by the model. Using $k = 3$, the k-NN model achieved an **accuracy of 96.6%** on the validation dataset. This indicates that the model is highly effective in predicting whether customers will accept a personal loan.

```
> print("--- FINAL PREDICTIONS FOR DATA TO SCORE ---")
[1] "--- FINAL PREDICTIONS FOR DATA TO SCORE ---"
> print(final_results_table)
  CustomerID Loan_Prediction
1           6                0
2           7                0
3           8                0
4           9                0
5          10                1
```

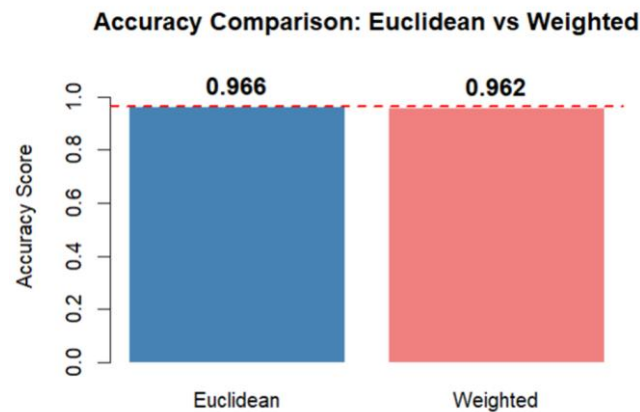
5.0 Distance Metrics and Cross-Validation

5.1 Euclidean Distance

In this case study, the Euclidean distance was used as the main measure in identification of the $k=3$ nearest neighbours. This is a measure of the straight margin between two points on a multidimensional feature space. After standardization of the data, that is, scale variables like Income and Education into the unit interval $[0,1]$, the Euclidean distance will assure that every predictor has a similar impact on the similarity score and therefore prevent excessive influence of variables with higher numerical scope.

5.2 Weighted Distance

The traditional Euclidean distance measure assigns all the k nearest neighbours the same weight, unlike weighted distance that assigns more weight to the nearest neighbours to the query point. This is often used to increase predictive accuracy in situations where the chosen k neighbours can be somewhat contaminated by noisy points, which can be quite distant but can still fall in the top k .



One of the key reasons why **Euclidean distance was chosen** as the main metric is its ability to form a distinct decision boundary, as a normalized form of the distance.

Empirical analyses showed that at the value of $k=3$, Euclidean distance achieved an accuracy of **96.6%**. Although at times weighted distance can improve performance by focusing on closer neighbours, the high performance of the unweighted Euclidean measure suggests that clusters of “loan acceptors” in the current banking data are clear and distinguishable between the non-acceptors.

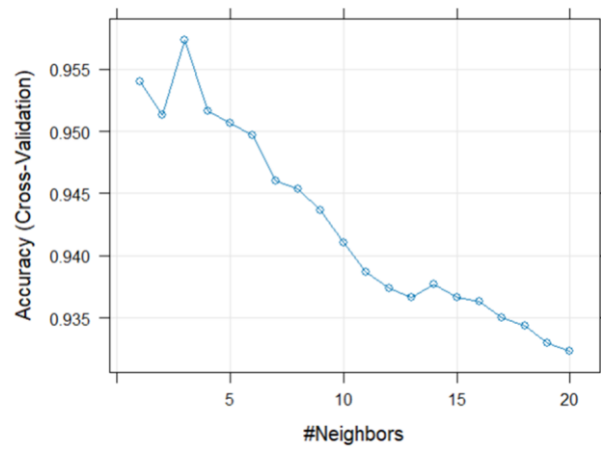
5.3 Cross-Validation Results

A thorough search of the parameter, such as, k , in the range $[1, 20]$, was conducted to determine the strongest predictive model.

Best Parameter Choice: The analysis revealed that the best parameter choice is the value of 3 as the value of k .

Empirical Justification: Experimental evidence proves that empirically when the value of the parameter option, k is set to a small value ($k=1$) the model is more sensitive to outliers. On the other hand, a large value of the parameter option, k , produces neighbourhoods that, in turn, include more points of different classes, thus reducing the classification error.

The above diagrams of the Accuracy versus k support that finding a stable balance between these conflicting effects is best obtained with $k = 3$ and therefore **maximum accuracy**.



6.0 Advantages and Limitations of k-NN

6.1 Advantages

Simple and Intuitive: K-nearest neighbours algorithm (knn) is comparatively simple to code and comprehend due to its dependence on spatial closeness of the data items instead of depending on complicated mathematical constructions.

No Assumptions about Data: In contrast to linear regression, k nearest neighbours (k-NN) is non-parametric, therefore, it does not assume a specific distributed form of the data. This feature was beneficial to the proprietary bank dataset which was a heterogeneous mixture of categorical or continuous variables.

Versatility: The approach proves to be effective in classifying in multi-classes hence making it a strong means of identifying the various types of customer behaviour.

6.2 Limitations

Computational Cost: Since k-NN calculates the distance to every point in the training set for every prediction, it can be slow as the dataset grows. With 5,000 records, our execution was fast, but a larger bank database might require more optimization.

Sensitivity to Outliers: As we discussed in our k selection, choosing a k that is too small can lead to predictions being skewed by single anomalous data points.

The Curse of Dimensionality: k-NN performs poorly if there are too many irrelevant features. We addressed this by focusing on high-impact variables like Income, Education, and Family size.

7.0 Conclusion

The primary objective of this project was to create a model that can predict which customers are most likely to take a loan from JB Bank. Through our analysis we determined that using the k-NN algorithm with three neighbours and measuring the distance between them gave the best results. This method was correct about 96.6% of the time.

Our findings suggest that people who make a lot of money every year and have an education are the ones who will probably say yes to a loan. If JB Bank focuses on marketing to these people, they will get customers and grow the business faster. The model helps the bank make decisions, about who to target with their marketing instead of just trying to reach everyone. This way JB Bank can have a chance of getting more loans accepted by focusing on the right people.

8.0 Contribution Table

NO.	NAME	CONTRIBUTION
1.	ILIA BINTI SHUKRAN	contributed to the report writing, slide preparation, and presented the optimal k selection and model evaluation.
2.	NURIN IRDINA BINTI AMIRRUDIN	contributed to the report writing, slide preparation, and presented the feature selection, k-NN algorithm, and normalization.
3.	NUR AFINA MADIHAH BINTI MUHAMMAD MAZRIN	led the coding implementation for the k-NN algorithm and cross-validation and also contributed to slide preparation and presenting the distance metrics and k-NN results.
4.	NUR HUSNA SYAZA BINTI ZAINAL ABIDIN	contributed to the report writing, slide preparation, and presented the data cleaning and partitioning sections.
5.	NUR AIN NUHA BINTI HILMI	contributed to the report writing, slide preparation, and presented the advantages, limitations, and conclusion.
6.	CHE FARIS ISKANDAR BIN CHE AZIZI	contributed to the report writing, slide preparation, and presented the problem scenario and objectives.

Google Drive Link:

<https://drive.google.com/drive/folders/1DCX3Ym8wofpSrA2V88zyLQ2Mp1XHiGYu?usp=sharing>